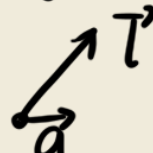


方向导数与梯度

 $\vec{l} = (\cos\alpha, \cos\beta, \cos\gamma)$

若 $\lim_{t \rightarrow 0^+} \frac{f(\vec{a} + t\vec{l}) - f(\vec{a})}{t}$ 存在

称作 f 在 $\vec{a}$ 处沿 $\vec{l}$ 的方向导数

例: $f(x, y, z) = x^2 + y^2 + z^2 + \sqrt{x^2 + y^2 + z^2}$

$$\frac{\partial f}{\partial x}(0, 0, 0) = \lim_{x \rightarrow 0} \frac{f(x, 0, 0) - f(0, 0, 0)}{x} = x + \frac{|x|}{x} \quad \text{极限不存在}$$

故偏导不存在

设 $\vec{l} = (\cos\alpha, \cos\beta, \cos\gamma)$

$$\frac{\partial f}{\partial \vec{l}}(0, 0, 0) = \lim_{t \rightarrow 0^+} \frac{f(t\cos\alpha, t\cos\beta, t\cos\gamma) - f(0, 0, 0)}{t} = \lim_{t \rightarrow 0^+} \frac{t^2 + t}{t} = 1$$

沿任意方向, 都存在方向导数

方向导数无法推出偏导数存在

$f(x, y, z)$ 可微, $\vec{l}$ 为单位向量, $\frac{\partial f}{\partial \vec{l}}$ 存在

$$\frac{\partial f}{\partial \vec{l}} = \left. \frac{d}{dt} \right|_{t=0} f(x + t\cos\alpha, y + t\cos\beta, z + t\cos\gamma)$$

链式法则 $= f'_x \cos\alpha + f'_y \cos\beta + f'_z \cos\gamma$

$$= (f'_x, f'_y, f'_z) \cdot \vec{l}$$

$$= \text{grad} f \cdot \vec{l} \quad (\text{梯度})$$

例 $f(x, y, z) = xyz + e^{xyz}$ 在 $(1, 2, 1)$ 处 沿 $\vec{l} = (2, -2, 1)$ 的方向导数

$$|\vec{l}| = \sqrt{2^2 + (-2)^2 + 1^2} = 3 \quad \frac{\vec{l}}{|\vec{l}|} = (\frac{2}{3}, -\frac{2}{3}, \frac{1}{3})$$

$$\frac{\partial f}{\partial x} = yz(1 + e^{xyz}) \big|_{(1,2,1)} = 2(1 + e^2)$$

$$\frac{\partial f}{\partial y} = xz(1 + e^{xyz}) \big|_{(1,2,1)} = 1 + e^2$$

$$\frac{\partial f}{\partial z} = xy(1 + e^{xyz}) \big|_{(1,2,1)} = 2(1 + e^2)$$

$$\frac{\partial f}{\partial l} \big|_{(1,2,1)} = (1 + e^2)(2, 1, 2) \cdot (\frac{2}{3}, -\frac{2}{3}, \frac{1}{3}) = \frac{4}{3}(1 + e^2)$$

梯度 $f(x, y, z)$ 可微 $(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z})$ 称为 f 的梯度. 记为 $\text{grad} f$. ∇f

$$\frac{\partial f}{\partial \vec{l}} = \nabla f \cdot \frac{\vec{l}}{|\vec{l}|} = |\nabla f| \cdot \cos \theta$$

$\vec{l}$ 与 梯度 导数 平行, f 的方向导数取 $\max = |\nabla f|$